

Project Report:

Security Analysis and Vulnerability Assessment

[Using the Open Bug Bounty Framework]

Project Title: Security Analysis and Vulnerability Assessment of drako.it and monroecountyhistory.org

Course Name: [CYBERSECURITY-CORIZO]

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✓ **Abstract**

This project demonstrates a systematic approach to web security testing on drako.it using the Open Bug Bounty platform. By combining reconnaissance, automated scanning, and manual testing, we aimed to identify and responsibly disclose vulnerabilities. Our findings reveal potential security weaknesses and provide remediation recommendations to improve the website's security posture.

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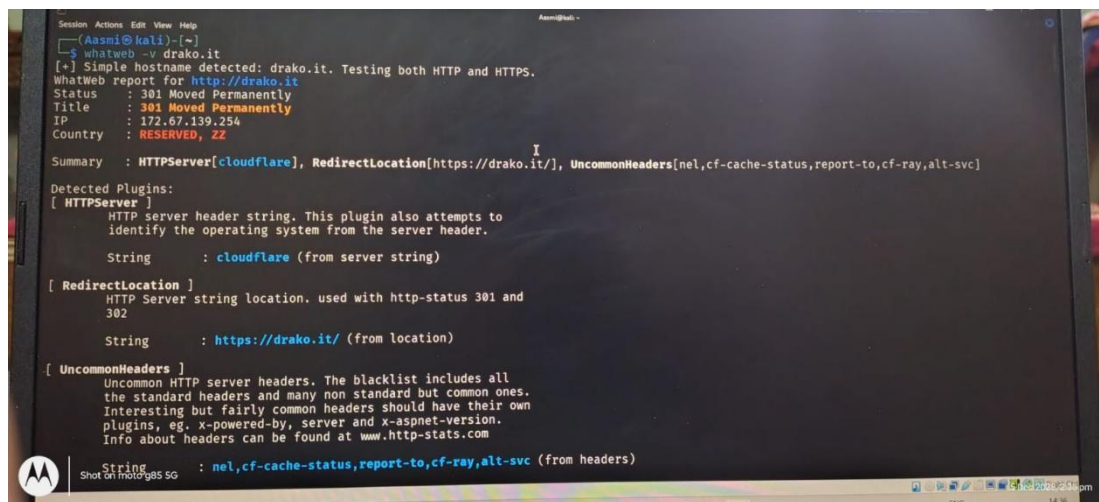
1. Introduction

Ensuring web application security is critical to protecting user data and maintaining trust. This project focuses on drako.it, a website selected for its manageable size and relevance for academic learning. Using Open Bug Bounty as a framework, we applied a structured testing methodology aimed at uncovering typical vulnerabilities and showcasing ethical security testing practices.

2. Testing Methodology

Our methodology was structured in phases:

- **Reconnaissance:** Identified technology stack, subdomains, and mapped key pages.
- **Automated Scanning:** Performed controlled scans using OWASP ZAP.
- **Manual Testing:** Validated automated findings and conducted deeper vulnerability checks such as SQL Injection, XSS, CSRF, and broken authentication.
- **Reporting:** Documented all confirmed issues and submitted reports responsibly via Open Bug Bounty.



```
Session Actions Edit View Help
(Aasmik@kali)-[~]
$ whatweb -v drako.it
[+] Simple hostname detected: drako.it. Testing both HTTP and HTTPS.
whatweb report for http://drako.it
Status : 301 Moved Permanently
Title : 301 Moved Permanently
IP : 172.67.139.254
Country : RESERVED, ZZ

Summary : HTTPServer[cloudflare], RedirectLocation[https://drako.it/], UncommonHeaders[nel,cf-cache-status,report-to,cf-ray,alt-svc]

Detected Plugins:
[ HTTPServer ]
HTTP Server header string. This plugin also attempts to
identify the operating system from the server header.
String : cloudflare (from server string)

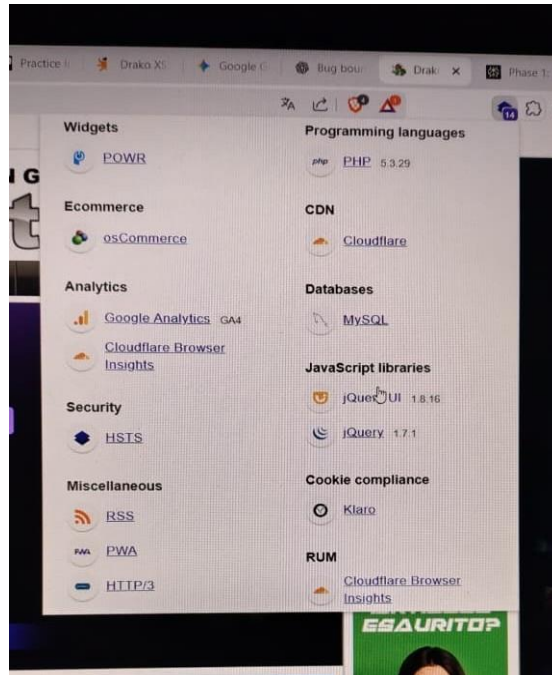
[ RedirectLocation ]
HTTP Server string location. used with http-status 301 and
302
String : https://drako.it/ (from location)

[ UncommonHeaders ]
Uncommon HTTP server headers. The blacklist includes all
the standard headers and many non standard but common ones.
Interesting but fairly common headers should have their own
plugins, eg. x-powered-by, server and x-aspnet-version.
Info about headers can be found at www.http-stats.com

String : nel,cf-cache-status,report-to,cf-ray,alt-svc (from headers)
```

3. Target Reconnaissance

- **Domain:** drako.it
- **Subdomains Found:** NA
- **Technology Stack:**
 1. Web Server: Information pending (typically Apache/Nginx)
 2. Programming Language: PHP
 3. CMS or Framework: Used Wappalyzer
 4. Mapping: Identified main pages, login panels, and input forms using tools like gobuster/dirb.



Wappalyzer Test

```
(Aasmik@kali)-[~]
$ gobuster dir -u https://monroecountyhistory.org -w /usr/share/wordlists/dirb/common.txt

Gobuster v3.8
by OJ Reeves (@TheColonial) & Christian Mehlmauer (@firefart)

[+] Url:             https://monroecountyhistory.org
[+] Method:          GET
[+] Threads:         10
[+] Wordlist:         /usr/share/wordlists/dirb/common.txt
[+] Negative Status codes: 404
[+] User Agent:      gobuster/3.8
[+] Timeout:         10s

Starting gobuster in directory enumeration mode

./hta (Status: 403) [Size: 199]
./git/HEAD (Status: 403) [Size: 199]
./htpasswd (Status: 403) [Size: 199]
./htaccess (Status: 403) [Size: 199]
/account (Status: 301) [Size: 248] [→ https://monroecountyhistory.org/account/]
/admin (Status: 301) [Size: 246] [→ https://monroecountyhistory.org/admin/]
/blog (Status: 301) [Size: 245] [→ https://monroecountyhistory.org/blog/]
/cart (Status: 301) [Size: 245] [→ https://monroecountyhistory.org/cart/]
/cgi-bin/ (Status: 403) [Size: 199]
/contact-us (Status: 301) [Size: 251] [→ https://monroecountyhistory.org/contact-us/]
/content (Status: 301) [Size: 248] [→ https://monroecountyhistory.org/content/]
/crossdomain.xml (Status: 200) [Size: 603]
/csv (Status: 301) [Size: 244] [→ https://monroecountyhistory.org/csv/]
/favicon.ico (Status: 200) [Size: 1150]
/feedback (Status: 301) [Size: 249] [→ https://monroecountyhistory.org/feedback/]
/fonts (Status: 301) [Size: 246] [→ https://monroecountyhistory.org/fonts/]
```

```
/account (Status: 301) [Size: 248] [→ https://monroecountyhistory.org/account/]
/admin (Status: 301) [Size: 246] [→ https://monroecountyhistory.org/admin/]
/blog (Status: 301) [Size: 245] [→ https://monroecountyhistory.org/blog/]
/cart (Status: 301) [Size: 245] [→ https://monroecountyhistory.org/cart/]
/cgi-bin/ (Status: 403) [Size: 199]
/contact-us (Status: 301) [Size: 251] [→ https://monroecountyhistory.org/contact-us/]
/content (Status: 301) [Size: 248] [→ https://monroecountyhistory.org/content/]
/crossdomain.xml (Status: 200) [Size: 603]
/csv (Status: 301) [Size: 244] [→ https://monroecountyhistory.org/csv/]
/favicon.ico (Status: 200) [Size: 1150]
/feedback (Status: 301) [Size: 249] [→ https://monroecountyhistory.org/feedback/]
/fonts (Status: 301) [Size: 246] [→ https://monroecountyhistory.org/fonts/]
/images (Status: 301) [Size: 247] [→ https://monroecountyhistory.org/images/]
/includes (Status: 301) [Size: 249] [→ https://monroecountyhistory.org/includes/]
/index (Status: 301) [Size: 246] [→ https://monroecountyhistory.org/index/]
/index.php (Status: 200) [Size: 33230]
/javascript (Status: 301) [Size: 251] [→ https://monroecountyhistory.org/javascript/]
/js (Status: 301) [Size: 243] [→ https://monroecountyhistory.org/js/]
/modules (Status: 301) [Size: 248] [→ https://monroecountyhistory.org/modules/]
/news (Status: 301) [Size: 245] [→ https://monroecountyhistory.org/news/]
/records (Status: 301) [Size: 248] [→ https://monroecountyhistory.org/records/]
/search (Status: 301) [Size: 247] [→ https://monroecountyhistory.org/search/]
/server-status (Status: 403) [Size: 199]
/tag (Status: 301) [Size: 244] [→ https://monroecountyhistory.org/tag/]
/uploads (Status: 301) [Size: 248] [→ https://monroecountyhistory.org/uploads/]
/vendor (Status: 301) [Size: 247] [→ https://monroecountyhistory.org/vendor/]

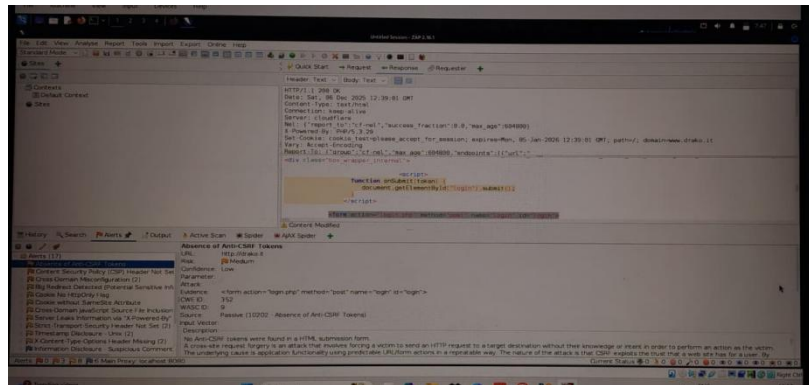
Progress: 4613 / 4613 (100.00%)
Finished
```

gobuster test

4. Automated Scanning Results

Using OWASP ZAP, we conducted a low-impact scan of drako.it. The scan detected several alerts ranging from informational to medium severity. After filtering false positives, the following were prioritized for deeper manual analysis:

1. Potential XSS in input fields
2. Security headers missing or incomplete
3. Possible outdated software components detected



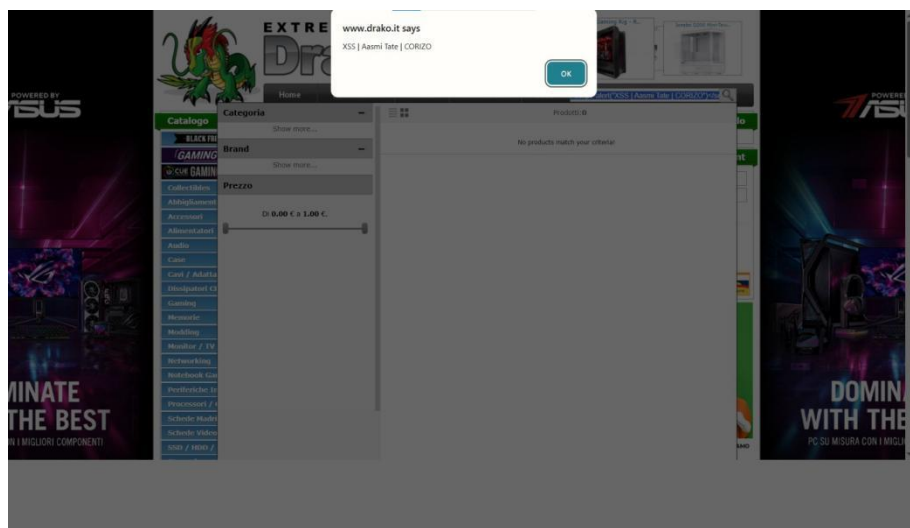
5. Manual Testing & Vulnerability Analysis

- **Vulnerability 1:** Reflected Cross-Site Scripting (XSS) in Search Parameter
- **Severity:** Medium
- **Description:** Input fields on the search page do not properly sanitize user input, allowing script injection.

Steps to Reproduce:

- i. Navigate to the search page on drako.it
- ii. Enter the payload `<script>alert('XSS')</script>` in the search input
- iii. Submit and observe the popup execution

- **Proof of Concept (PoC):**



- iv. Potential Impact: An attacker could execute malicious scripts, stealing cookies or hijacking user sessions.
- v. Remediation: Implement input validation and output encoding to sanitize inputs.

6. Conclusion & Lessons Learned

This project reinforced the value of a methodical security testing approach combining automated and manual techniques. We learned the importance of understanding a website's architecture before testing and saw firsthand common vulnerabilities on real-world targets like drako.it, monroecountyhistory.org. Responsible reporting helps improve overall web ecosystem security.

7. References

- ✧ OWASP ZAP: <https://www.zaproxy.org/>
- ✧ Open Bug Bounty: <https://www.openbugbounty.org/>
- ✧ Wappalyzer: <https://www.wappalyzer.com/>
- ✧ gobuster: <https://github.com/OJ/gobuster>